Gamma Distribution

1 Definition

A continuous positive random variable X is said to have a Gamma distribution if its PDF is given by,

$$f(x) = \begin{cases} \dfrac{e^{-x/\theta}x^{n-1}}{\theta^n \ \Gamma n} & ,0 < x < \infty, n > 0 \\ 0 & , \text{ otherwise} \end{cases}$$

Discussion:

$$\Gamma n = (n-1)\Gamma n - 1$$
$$\Gamma n = (n-1)!, \text{ if } n \in \mathbb{N} \text{ (integer)}$$
$$\Gamma(\frac{1}{2}) = \sqrt{\pi}$$

As, $\int_0^\infty e^{-x}x^{n-1}dx = \Gamma n$

$$\begin{aligned} \int_0^\infty e^{-x/\theta}x^{n-1}dx &= \int_0^\infty e^{-z}(\theta z)^{n-1}\theta dz \quad , \tfrac{x}{\theta} = z \Rightarrow x = z\theta \\ &= \theta^n \int_0^\infty e^z z^{n-1}dz \qquad\qquad \Rightarrow dx = \theta dz \\ &= \theta^n \Gamma n \\ \Rightarrow \frac{\int_0^\infty e^{-x/\theta}x^{n-1}dx}{\theta^n \Gamma n} &= 1 \end{aligned}$$

Another format, $\int_0^\infty e^{-\theta x}x^{n-1}dx = \int_0^\infty e^{-z}\left(\frac{z}{\theta}\right)^{n-1}\frac{dz}{\theta}, \quad \theta x = z$

$$\begin{aligned} \int_0^\infty e^{-\theta x}x^{n-1}dx &= \int_0^\infty e^{-z}\left(\frac{z}{\theta}\right)^{n-1}\frac{dz}{\theta}, \quad \theta x = z \\ &= \frac{1}{\theta^n}\int_0^\infty e^{-z}z^{n-1}dz \\ &= \frac{\sqrt{n}}{\theta^n} \\ \Rightarrow \frac{\theta^n \int_0^\infty e^{-\theta x}x^{n-1}dx}{\Gamma n} &= 1 \end{aligned}$$

So, we can also write it as, $f(x) = \begin{cases} \dfrac{\theta^n e^{-\theta x}x^{n-1}}{\Gamma n}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$

Notation:

$X \sim \text{Gamma}(\theta, n)$ (This is not an universal notation)

2 Moments

$$\begin{aligned} \mu_r' = E(X^r) &= \frac{1}{\theta^{n\Gamma n}}\int_0^\infty x^r e^{-x/\theta}x^{n-1}dx \\ &= \frac{1}{\theta^{n\Gamma n}}\int_0^\infty e^{-x/\theta}x^{r+n-1}dx \end{aligned}$$

$$= \frac{\theta^{n+r}\Gamma(n+r)}{\theta^n\sqrt{n}} \text{ if } \quad r > -n$$
$$= \frac{\theta^r\Gamma(n+r)}{\Gamma n}$$

$$E(X) = \mu_1' = \frac{\theta\Gamma(n+1)}{\Gamma n} = \theta n$$
$$E\left(X^2\right) = \mu_2' = \frac{\theta^2\Gamma(n+2)}{\Gamma n} = \theta^2(n+1)n$$
$$V\left(X^2\right) = E\left(X^2\right) - E^2(X)$$
$$= \theta^2 n(n+1) - \theta^2 n^2$$
$$= \theta^2 n(n+1-n)$$
$$= \theta^2 n$$

3 Harmonic Mean

$$\frac{1}{H} = E\left(\frac{1}{X}\right) = \frac{\theta^{-1}\Gamma(n-1)}{\Gamma n} = \frac{1}{\theta(n-1)}$$
$$\Rightarrow H = \theta(n-1), \text{ if } n > 1$$

4 Mode

The value of X for which the PDF is maximum.
For $0 < x < \infty$,

$$\ln f_X(x) = -\frac{x}{\theta} + (n-1)\ln x - \underbrace{\ln\left(\theta^n\sqrt{n}\right)}_{\text{constant}}$$
$$\frac{\partial}{\partial x}\ln f_X(x) = -\frac{1}{\theta} + \frac{(n-1)}{x}$$
$$= \frac{(n-1)\theta - x}{\theta x} \begin{cases} > 0, x < (n-1)\theta \\ < 0, x > (n-1)\theta \end{cases}$$

$\ln f_x(x)/f_x(x)$ first increases, then attains it's maxima at $x = (n-1)\theta$, then finally decreases. Hence, $x = (n-1)\theta, n > 1$ is the mode.

5 Skewness

$$S_K = \frac{\text{mean } - \text{ mode}}{\text{sd}}$$
$$= \frac{\theta n - (n-1)\theta}{\sqrt{\theta^2 n}}$$
$$= \frac{\theta}{\theta\sqrt{n}} = \frac{1}{\sqrt{n}} > 0$$

$S_K \to 0$ as $n \to \infty$, the gamma distribution is positively skewed for $n > 1$.

6 Problems

1. Show that, $\int_t^\infty e^{-x/\theta}x^{p-1}dx < \frac{\theta^{p+1}\Gamma(p+1)}{t}$.

$$\Rightarrow \int_t^\infty \frac{e^{-x/\theta}x^{p-1}dx}{\theta^p\Gamma p} < \frac{\theta p}{t}; \qquad \sqrt{p+1} = p! = p(p-1)! = p\sqrt{p}$$

Let $X \sim \text{Gamma}(\theta, p)$

$$\begin{aligned} P(X > t) &\leqslant \frac{E(X)}{t}, \text{ using Morkov's inequality as } P(X > 0) = 1 \\ &\leqslant \frac{\theta p}{t} \\ \Rightarrow \int_t^\infty \frac{e^{-x/\theta} x^{p-1}}{\theta^p \Gamma p} dx &< \frac{\theta p}{t} \\ \Rightarrow \int_t^\infty e^{-x/\theta} x^{p-1} dx &< \frac{\theta^{p+1} \Gamma(p+1)}{t} \end{aligned}$$

2. If $X \sim \text{Gamma}(\theta, n)$, comment of the skewness and hence argue that $P(X > E(X)) < \frac{1}{2}$.

Markov's inequality will give us a trivial bound. Here, X is positively skewed as $E(X) = n\theta$ and mode $= (n-1)\theta$.

$$P(X > \text{ mean }) < P(X > \text{ median }) = \frac{1}{2}$$

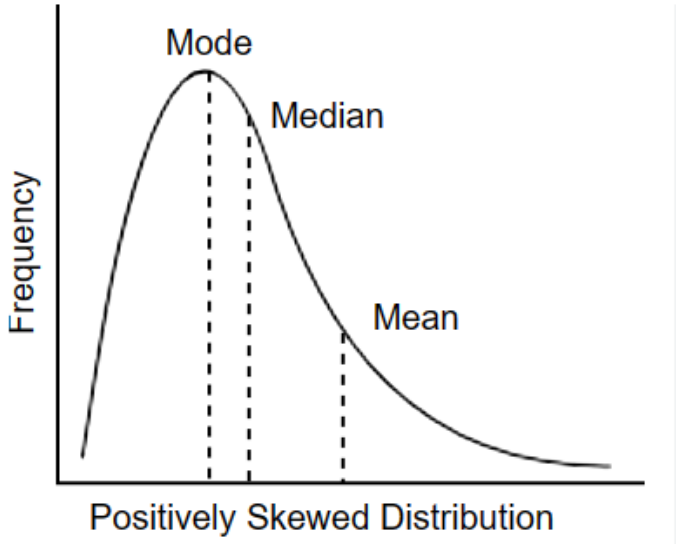


Figure 1: Plot is taken from here

7 Standard Gamma Distribution

$$f(x) = \begin{cases} \dfrac{e^{-x} x^{n-1}}{\Gamma n}, & 0 < x < \infty \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = n, \quad \text{var}(X) = n.$$

CDF:

$$\begin{aligned} F(x) = \int_{-\infty}^x f(t)dt &= \int_{-\infty}^0 \underbrace{f(t)dt}_{=0} + \int_0^x f(t)dt \\ &= \int_0^x \frac{e^{-t} t^{n-1}}{\Gamma n} dt \\ &= \frac{\int_0^x e^{-t} t^{n-1} dt}{\int_0^\infty e^{-t} t^{n-1} dt} \end{aligned}$$

which is known as **Incomplete Gamma function** and denoted by $\Gamma x^{(n)}$ (Gamma uptill x of n).

Exact evaluation of $F(x) = \Gamma x^{(n)}$ is not possible for all $n > 0$ but numerical methods can be applied. The values of $\Gamma x^{(n)}$ has been extensively tabulated for different values of x by Karl Pearson.

The calculation of cumulative probabilities of Poisson distribution is faciliated by the use of Incomplete Gamma Function.